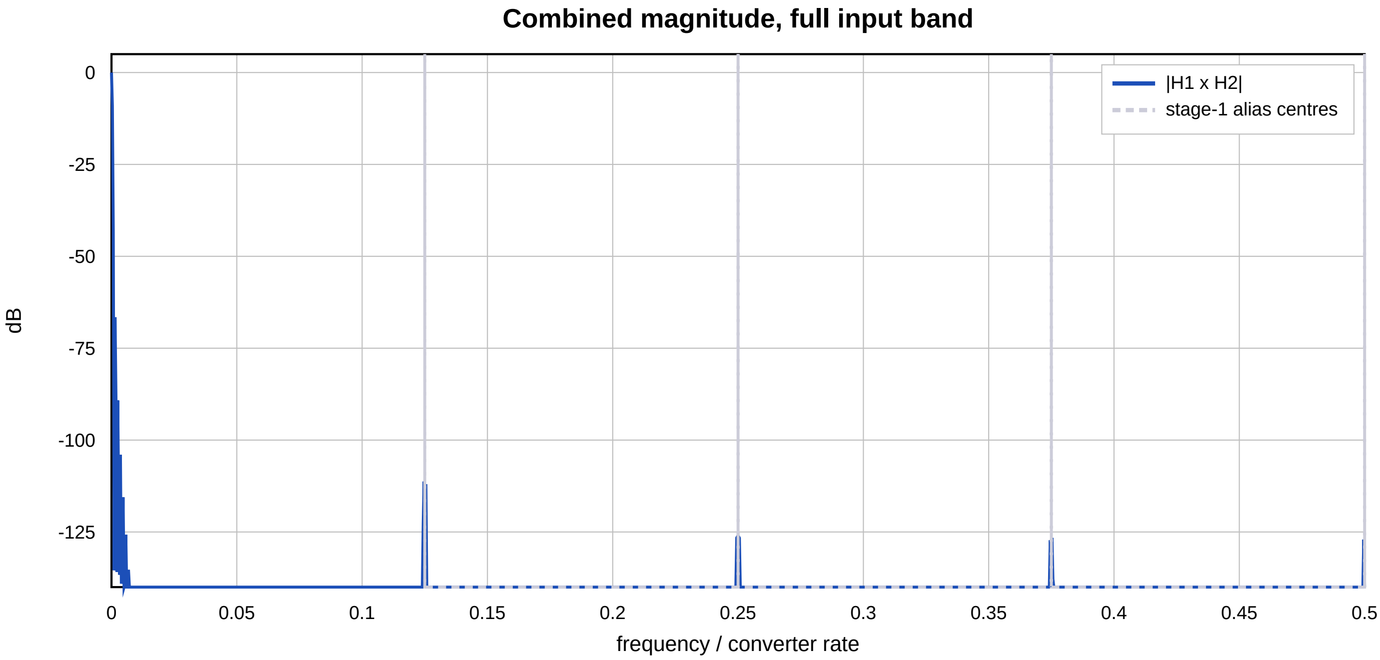
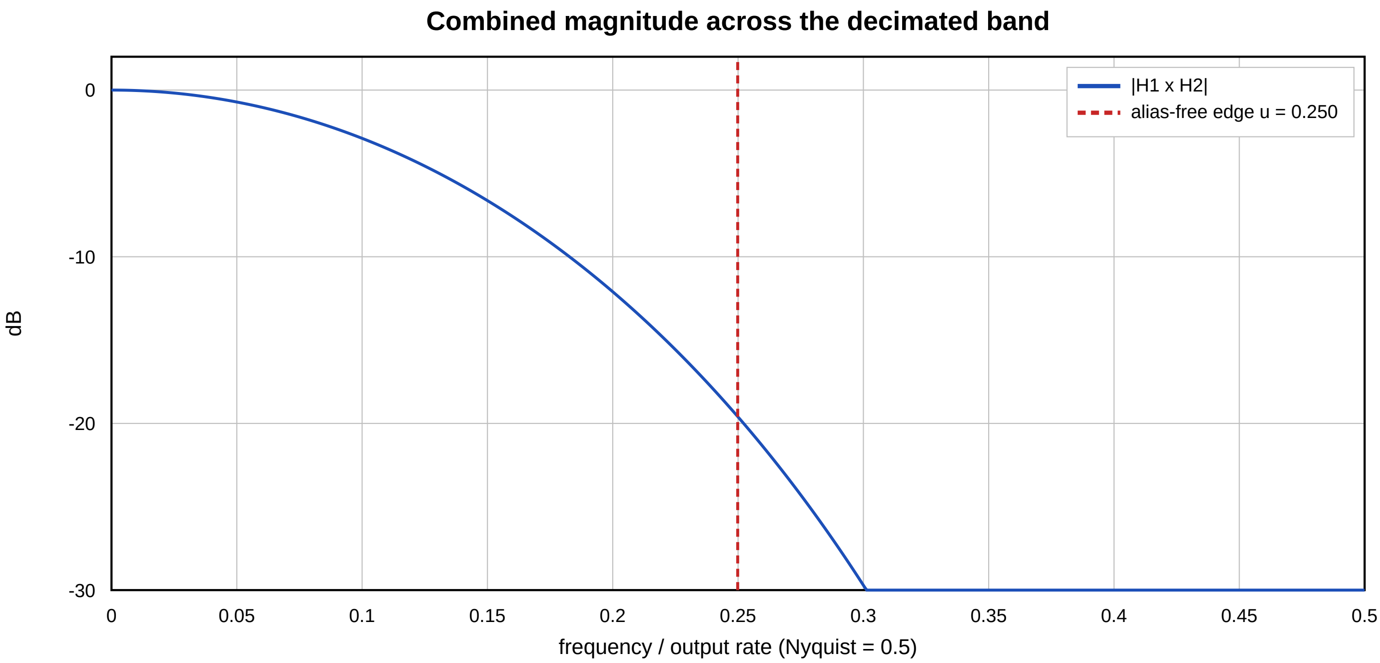
Decimation Chain - Derived Design

125.000 MHz / 488 = 256.1475 kHz, split [8, 61], 64.000 kHz alias-free





Stages

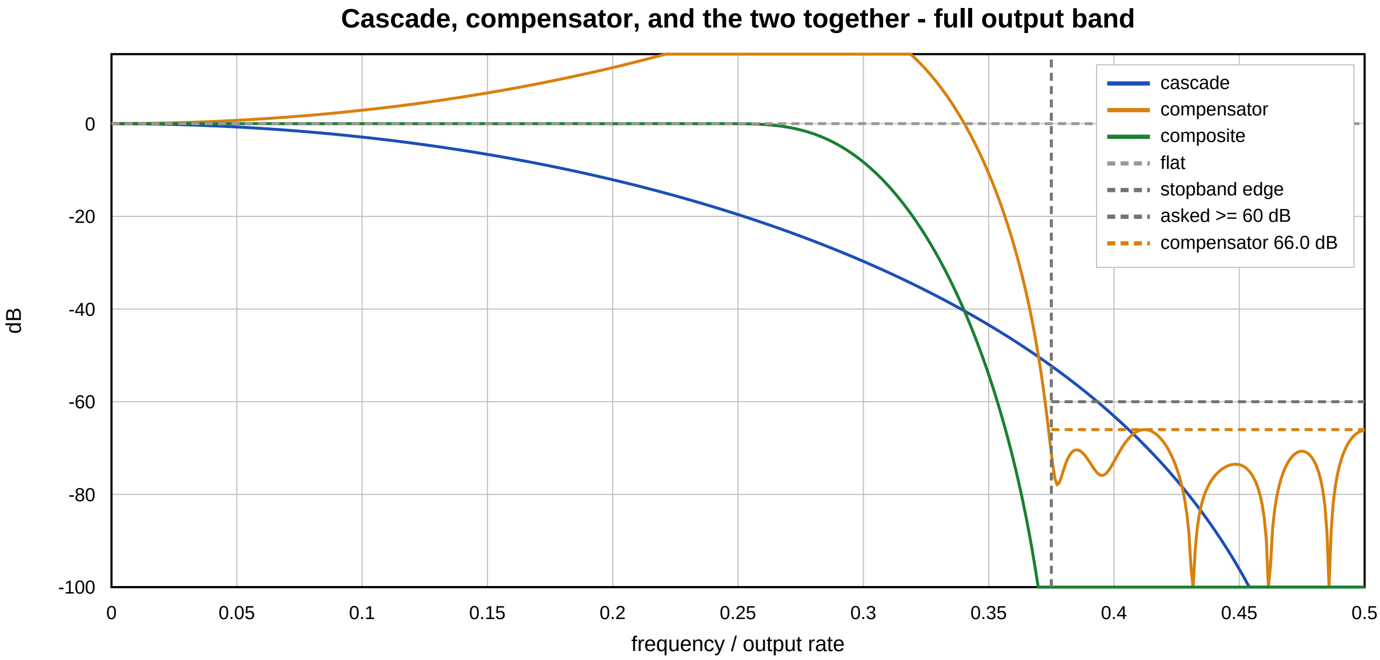
stage 1: /8 N=2 M=1 at 125.000 MHz
  accumulator 22 bits, prune budget 12, widths [16, 16, 16, 16]
  64 register bits, in 16 bits out 16 bits

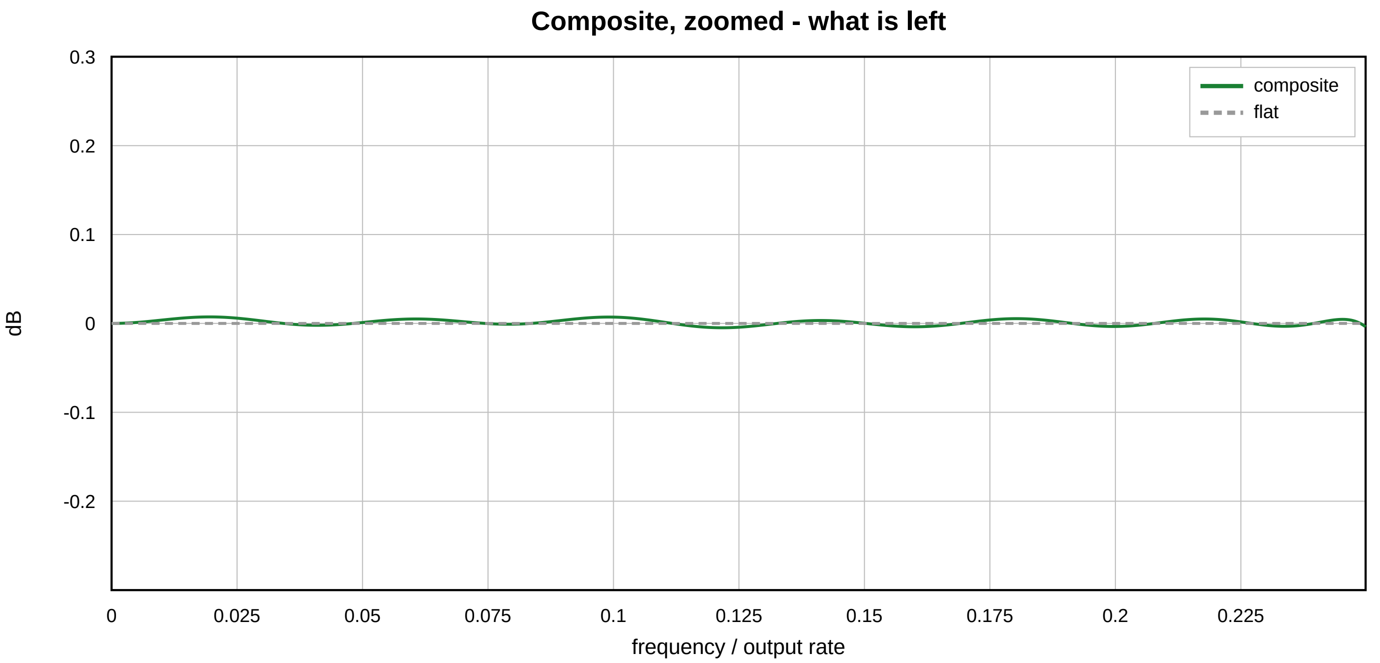
stage 2: /61 N=5 M=2 at 15.625 MHz
  accumulator 51 bits, prune budget 47, widths [37, 30, 24, 18, 16, 16, 16, 16, 16, 16]
  205 register bits, in 16 bits out 16 bits

A stage's nulls must reject the aliases of its own decimation: once energy has folded into the band no later stage can remove it, so every stage carries the full requirement.

Compensation and Result

47 taps, 16-bit coefficients, 24 multipliers





Achieved

ripple .... 0.0122 dB (asked <= 0.100)
alias rejection ... 67.3 dB   (asked >= 60.0)
output SNR .... 84.6 dB   (asked >= 80.0)
stopband .... 66.0 dB above 0.75 Nyquist (asked >= 60.0)
compensator alone; the composite is lower by the cascade's rolloff

register bits ..... 269 (rate-weighted cost 89.6)
taps .... [-5, -12, 31, 29, -123, 18, 298, -294, -399, 931, 6, -1727, 1370, 1890, -3735, -253, 6106, -3938, -6681, 10266, 3975, -17439, 1206, 25152, 1206, -17439, 3975, 10266, -6681, -3938, 6106, -253, -3735, 1890, 1370, -1727, 6, 931, -399, -294, 298, 18, -123, 29, 31, -12, -5]
DC gain .... 1.000000 (exact by construction)
alternative .... [4, 61, 2], cost 96.8, 384 bits - feasible, but costlier by the rate-weighted model

Rate-weighted cost is a proxy, not an area figure: flops cost the same however slowly they are clocked. It captures that a wide adder at the converter rate is the hard part.